Dynamic Job Shop & Flow Shop Scheduling Engine

Operator & User Mini Handbook — Complete Guide to Every Button, Metric, Engine & Algorithm

1. System Overview: What Is This Project?

This platform is an **AI-Powered Real-Time Dynamic Job Shop and Flow Shop Production Scheduler**. In modern manufacturing plants (like heavy railway engineering, automotive foundries, and CNC centers), multiple custom work orders compete for the same physical machines. If machines are scheduled poorly, work-in-progress (WIP) queues pile up, jobs get delayed, and machines sit idle.
This system solves that by offering: **1) Real-Time Simulation**, **2) Exact Optimization via Google OR-Tools CP-SAT**, **3) Academic Flow Shop Heuristics (Palmer, CDS, NEH)**, **4) Deep Learning (Pointer Networks)**, and **5) Machine Learning Strategy Recommendation**.

2. Top Navigation Bar: What Every Button Means

Control / Button	Visual Icon	What It Actually Does / Purpose
Sim Clock Display	*e.g. 540 hrs*	Shows the current virtual factory time in **hours**. All operations and progress move relative to this clock.
+5, +15, +30, +60 hrs	*FastForward (Blue)*	Advances virtual factory time. Machines finish active tasks, IDLE machines pick the next queued job, and telemetry updates.
Auto Play (1x, 2x, 5x)	*Play / Pause (Green)*	Hands-free execution. Runs the factory continuously at selected speeds so you can watch live progress on the Gantt chart.
Export CSV	*Download (Cyan)*	Exports the exact current schedule, including job dispatches, assigned machines, start and end hours, into an Excel-ready CSV file.
Upload Excel/CSV	*Upload (Blue)*	Batch registers multiple jobs at once from an Excel (.xlsx) or CSV file. Automatically reads machines, operations, and due dates.
Add Dynamic Job	*PlusCircle (Emerald)*	Opens a form to add a single emergency or custom job. Automatically determines job priority from due date slack.
Reset Demo	*RotateCcw (Slate)*	Restores the factory to the standard 10 baseline demo jobs and resets the clock to 0 hrs without clearing algorithms.
Clear All	*Trash (Red)*	Clears all jobs from the factory floor to simulate a clean empty shop floor ready for custom input.
How It Works	*HelpCircle (Cyan)*	Launches an interactive 5-step tutorial walkthrough highlighting each major dashboard section.

3. Factory KPI Metric Cards: What Each Card Means

KPI Card Title	Unit	Industrial Meaning & Calculation
Makespan (Cmax)	*Hours (hrs)*	Total elapsed time until the very last operation of the last job is completed. Lower is better.
Total Tardiness	*Hours (hrs)*	Sum of delivery delay past the due date: $\sum \max(0, C_j - DD_j)$. Zero means all orders delivered on time.
Late Jobs	*Count*	Number of customer orders that missed their promised deadline ($C_j > DD_j$).
Avg Waiting Time	*Hours (hrs)*	Average time jobs spend queueing in front of machines before getting worked on.
Schedule Utilization	*Percentage (%)*	Percentage of total shop floor machine capacity actively engaged in cutting, moulding, or casting.
Solver Status / Latency	*Status / ms*	Displays whether the solution is OPTIMAL or FEASIBLE and execution time in milliseconds (typically 0.2 to 3 ms).

4. Production Job Lifecycle Dashboard & Table Actions

This interactive table lists every active job order on the shop floor with its status, progress, assigned machine, and deadlines.

Action Column Icons:
• ═■ **Pencil Icon (Edit Job):** Appears **only for Unstarted / Pending jobs** waiting in the queue. Once a job starts cutting on a machine or finishes completely, industrial protocol prevents altering its physical work order. Clicking this icon opens a modal to adjust due dates, stage names, or machine durations.
• ■■ **Trash Icon (Delete Job):** Cancels and removes an unstarted job order and immediately recalculates the schedule.

Automatic Priority Badges:
The system calculates priority automatically from Due Date Slack ($Slack = DD - [Arrival + \sum Duration]$):
• **P1 (Critical / Urgent):** Assigned when $Slack \le 5\text{ hrs}$. Requires immediate machine allocation.
• **P2 (High Priority):** Assigned when $5 < Slack \le 20\text{ hrs}$.
• **P3 (Normal / Routine):** Assigned when $Slack > 20\text{ hrs}$. Ample buffer time.

5. 10-Machine Real-Time Factory Floor Telemetry

Ten production stations ($M_1 \dots M_{10}$) represent physical primary foundry stages and parallel auxiliary stations:
• **M1:** Pattern Making | **M2:** Mould Making | **M3:** Casting (Foundry) | **M4:** Heat Treatment | **M5:** Machining Center
• **M6-M10:** Auxiliary parallel prep, dry, pouring, tempering, and quality finishing centers.
• **'Sim Breakdown (+20 hrs)':** Simulates a real-world machine outage (e.g., motor failure). The engine marks the machine **DOWN** in red, suspends its progress, and instantly re-routes pending work to available parallel machines without deadlock.
• **'Recover Machine':** Simulates completed maintenance, returning the station to **IDLE**.

6. AI Strategy Advisor: How Suggestions Work & Where Palmer Is

Where to find Palmer's Algorithm:
In the **'AI Strategy Advisor & Dispatch'** panel on the right side of the screen, look at the dropdown menu under **'Active Scheduling Engine'**. Select **'Palmer's Heuristic Algorithm (Slope Index)'** and click the blue **'Re-Solve'** button. The engine will instantly compute Palmer's slope indices for all jobs and dispatch them in optimal descending order.

How the ML Strategy Recommendation Works:
The AI Advisor continuously analyzes shop floor conditions (workload urgency, machine congestion, queue variance, breakdown counts) and uses a trained Random Forest model to recommend the best dispatching rule:
• **SPT:** Recommended during high backlogs to minimize average queue waiting.
• **EDD:** Recommended when deadlines are tight to eliminate late penalty costs.
• **AGING:** Recommended when older orders sit too long in queue to prevent starvation.
• **PALMER:** Recommended for multi-stage continuous flow patterns.
• **CP-SAT:** Recommended for finding the mathematically provable minimum makespan.
Clicking **'■ Apply Recommendation & Re-Solve'** switches to the recommended rule in one click.

7. Fair Algorithm Benchmark & Performance Matrix: What It Shows

Located at the bottom of the dashboard, this table takes the **exact current factory workload** and runs all seven scheduling engines in parallel side-by-side to show you how each one performs:

Algorithm in Matrix	Type	Optimization Logic / Core Behavior
FCFS	*Baseline Heuristic*	Dispatches jobs strictly in the order they arrived on the shop floor.
SPT	*Heuristic*	Dispatches shortest duration jobs first; minimizes average waiting time.
EDD	*Heuristic*	Dispatches jobs with the earliest due date first; minimizes maximum tardiness.
CR (Critical Ratio)	*Heuristic*	Considers remaining slack relative to work content; balances urgency dynamically.
AGING	*Heuristic*	Anti-starvation priority rule; guarantees older jobs are not trapped behind quick jobs.
PALMER	*Flow Shop Heuristic*	Computes slope index $S_j = \sum (2i - m - 1)p_{j,i}$ to optimize continuous flow shops.
OR-TOOLS CP-SAT	*Exact Solver*	Google Constraint Programming solver guaranteeing the absolute global minimum makespan.

8. Quick 5-Step Operator Walkthrough

Step 1: Check Current Shop Floor State — Look at the top KPI cards (Makespan, Tardiness, Late Jobs).
Step 2: Add or Upload Work Orders — Use **'Upload Excel/CSV'** to batch load test cases, or click **'Add Dynamic Job'** for single jobs.
Step 3: Modify Pending Orders — Click the ✏ **Pencil icon** next to any unstarted job in the table to modify its parameters.
Step 4: Select an Algorithm & Re-Solve — Choose **Palmer**, **CP-SAT**, **SPT**, or **EDD** from the dropdown and click **'Re-Solve'**.
Step 5: Advance Time & Watch Execution — Click **'+15 hrs'** or toggle **'Auto Play'** to watch the Gantt chart and machine stations execute tasks live.

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GitHub:	https://github.com/abhisheksaiguntuku/DYNAMIC-JOB-SCHEDULER